

# SYED TALAL WASIM

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## EDUCATION

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### University of Bonn

*Ph.D. in Computer Vision*

Jan 2024 – Ongoing

Bonn, Germany

- Supervisor: Professor Juergen Gall — **6 publications at top-tier venues (CVPR, ICCV, ICLR)**
- Researching long-term multimodal image and video understanding, world modeling, and efficient architectures
- Designed Video-Panda (CVPR 2025) and GroupMamba (CVPR 2025) to advance parameter-efficient visual modeling, achieving  $6.5\times$  encoder size reduction and 83.3% top-1 on ImageNet-1K with 26% fewer parameters, respectively
- Pushed long-term activity anticipation with MixANT (ICCV 2025), using dynamic forget-gate selection via mixture-of-experts in SSMS, outperforming state-of-the-art on 50Salads, Breakfast, and Assembly101
- Built STING-BEE (CVPR 2025), the first VLM for X-ray baggage security, trained on a self-curated 46,642-sample dataset across 21 threat categories with state-of-the-art cross-domain generalization
- Trained RedSage (ICLR 2026), an open-source cybersecurity LLM on 11.8B tokens, surpassing baselines by up to +5.59 points on cybersecurity benchmarks and +5.05 points on general LLM benchmarks

### Université de Bordeaux

*MS Image Processing and Computer Vision*

Sep 2019 – Jun 2021

Bordeaux, France

- MS Thesis: “Automatic Typography Analysis on Figurative Content”
- Funded by the Erasmus Mundus Joint Masters Degree (EMJMD) Scholarship Program
- First semester spent at Pázmány Péter Catholic University in Hungary, and second semester at the Universidad Autónoma de Madrid in Spain through the Erasmus mobility program

### Computer Vision Lab, EPFL

*MS Thesis Intern*

Feb 2021 – Jul 2021

Lausanne, Switzerland

- Supervisor: Dr. Mathieu Salzmann
- Conducted thesis research on typography analysis using Transformers for both RGB and Vector images
- Published findings in the Journal of Data Mining in Digital Humanities

### Habib University

*BS Electrical Engineering - Minor in Computer Science*

Sep 2015 – Jun 2019

Karachi, Pakistan

- BS Thesis: “SquadBot: A Multi-Agent Robotics Teaching and Research Platform”
- Dean’s and President’s Honor List (All 8 semesters)
- Awarded Merit Scholarship (65% tuition) and High Academic Achievement Scholarship (10% additional per semester based on performance)
- Graduated with Dean’s Medal (1<sup>st</sup> position in program) and Best Thesis Award

### Stanford University

*Summer International Honors Exchange Program*

Jun 2017 – Aug 2017

Stanford, USA

- Coursework: Technology Entrepreneurship, Leading Trends in IT, Smart Cities & Communities

## WORK EXPERIENCE

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### Mohamed Bin Zayed University of AI

*Research Associate*

Apr 2022 – Dec 2023

Masdar City, Abu Dhabi, UAE

- Supervisor: Professor Salman Khan — **6 publications at top-tier venues (CVPR, ICCV, NeurIPS)**
- Led research on multimodal video representation learning and out-of-distribution generalization
- Designed Video-FocalNets (ICCV 2023), achieving state-of-the-art video recognition on five benchmarks at lower computational cost than transformer-based methods

- Unified supervised and zero-shot video classification under a single training scheme through Vita-CLIP (CVPR 2023), achieving state-of-the-art zero-shot results on Kinetics-600, HMDB51, and UCF101
- Created VideoGrounding-DINO (CVPR 2024), the first open-vocabulary spatio-temporal video grounding model, outperforming prior methods by 4.88 m\_vIoU on HC-STVG
- Boosted hardware reliability of CLIP-based classifiers by  $5.5\times$  (NeurIPS 2023) and cut GPU memory for long-video segmentation by 87% via MAVOS (WACV 2025)

## Empathic Computing Lab, University of South Australia

Jul 2020 – Mar 2021

### Research Intern

*Remote (due to covid-19)*

- Supervisor: Professor Dr. Mark Billingham
- Researched multimodal emotion recognition by integrating facial micro-expressions, EEG, GSR, and PPG signals for improved affect detection
- Published findings in Frontiers in Psychology, demonstrating that involuntary micro-expressions provide more reliable emotional indicators than macro-expressions

## SKILLS

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**Research Areas:** Video Understanding, Large Language Models, Vision-Language Models, Efficient Neural Architectures, Multimodal Learning, Representation Learning

**Programming:** Python (Advanced), C/C++ (basic), C# (basic), Java (Basic)

**Common ML Tools:** Pandas, Numpy, Scikit-Learn, Tensorflow/Keras, OpenCV, Pytorch

**AR/VR and Game Engines:** Unity 3D, HoloLens 1

**Languages:** English: C2 (Expert), Urdu: Native

## ACADEMIC SERVICES

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- **Conference/Journal Reviewers:** CVPR, ICCV, ECCV, WACV, ACCV, NeurIPS, ICLR, ICML, AAAI, TPAMI, TNNLS, TIP, TMLR, IJCV, Pattern Recognition
- **Outstanding Reviewer at NeurIPS**

## AWARDS and GRANTS

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### Ph.D.

- **Compute Award (Jupiter Supercomputer):** 54.0 million GPU compute hours for Holistic Multi-modal Egocentric Video Forecasting
- **Compute Award (Leonardo Supercomputer):** 1.6 million GPU compute hours for Encoder-Free Video Language Models
- **Compute Award (Leonardo Supercomputer):** 3.0 million GPU compute hours for Large-Scale Open-Vocabulary Video Understanding and Anticipation
- **Compute Award (Leonardo Supercomputer):** 2.5 million GPU compute hours for Large-Scale Robust Vision Language Models
- **Winter School Grant:** MENA Winter School on Machine Learning 2025
- **Winter School Grant:** ELLIS Winter School on Foundational Models 2024

### Masters

- **Erasmus Mundus Scholarship:** Two year fully funded scholarship for MS studies
- **Summer School Grant:** ETH Zurich Robotics Summer School and Symposium 2021

# PUBLICATIONS AND PATENTS

## CONFERENCES

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1. N. Suryanto, M. Naseer, P. Li, **S. T. Wasim**, J. Yi, J. Gall, P. Ceravolo, and E. Damiani, “RedSage: A cybersecurity generalist LLM,” in *ICLR*, 2026
2. **S. T. Wasim**, H. Suleman, O. Zatsarynna, M. Naseer, and J. Gall, “MixANT: Observation-dependent memory propagation for stochastic dense action anticipation,” in *ICCV*, 2025
3. J. Yi, **S. T. Wasim**, Y. Luo, M. Naseer, and J. Gall, “Video-Panda: Parameter-efficient alignment for encoder-free video-language models,” in *CVPR*, 2025
4. A. Shaker, **S. T. Wasim**, S. Khan, J. Gall, and F. Khan, “Groupmamba: Parameter-efficient and accurate group visual state space model,” in *CVPR*, 2025
5. D. Velayudhan, A. Ahmed, M. Alansari, N. Gour, A. Behouch, T. Hassan, **S. T. Wasim**, N. Maalej, M. Naseer, J. Gall, M. Bennamoun, E. Damiani, and N. Werghi, “STING-BEE: Towards vision-language model for real-world x-ray baggage security inspection,” in *CVPR*, 2025
6. A. Shaker, **S. T. Wasim**, M. Danelljan, S. Khan, M.-H. Yang, and F. Khan, “Efficient video object segmentation via modulated cross-attention memory,” in *WACV*, 2025
7. **S. T. Wasim**, M. Naseer, S. Khan, M.-H. Yang, and F. Khan, “VideoGrounding-DINO: Towards open-vocabulary spatio-temporal video grounding,” in *CVPR*, 2024
8. **S. T. Wasim**, K. H. Soboka, A. Mahmoud, S. Khan, D. Brooks, and G.-Y. Wei, “Hardware resilience properties of text-guided image classifiers,” in *NeurIPS*, 2023
9. **S. T. Wasim**, M. U. Khattak, M. Naseer, S. Khan, M. Shah, and F. Khan, “Video-FocalNets: Spatio-temporal focal modulation for video action recognition,” in *ICCV*, 2023
10. M. U. Khattak, **S. T. Wasim**, M. Naseer, S. Khan, M.-H. Yang, and F. S. Khan, “Self-regulating prompts: Foundational model adaptation without forgetting,” in *ICCV*, 2023
11. **S. T. Wasim**, M. Naseer, S. Khan, F. Khan, and M. Shah, “Vita-CLIP: Video and text adaptive clip via multimodal prompting,” in *CVPR*, 2023

## JOURNALS

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1. H. Suleman, **S. T. Wasim**, M. Naseer, and J. Gall, “StableMamba: Distillation-free scaling of large ssms for images and videos,” *International Journal of Computer Vision (IJCV)*, 2026
2. **S. T. Wasim**, R. Collaud, L. Défayes, N. Henchoz, M. Salzmann, and D. Ribes, “Toward automatic typography analysis: serif classification and font similarities,” *Journal of Data Mining in Digital Humanities (JDMDH)*, 2023
3. N. Saffaryazdi, **S. T. Wasim**, K. Dileep, A. F. Nia, S. Nanayakkara, E. Broadbent, and M. Billingham, “Using facial micro-expressions in combination with eeg and physiological signals for emotion recognition,” *Frontiers in Psychology*, 2022

## PATENTS

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1. **S. T. Wasim**, M. U. Khattak, M. Naseer, S. Khan, and F. Khan, “System and method for modeling local and global spatio-temporal context in video for video recognition,” U.S. Patent 18411928, Jul. 2025
2. M. U. Khattak, **S. T. Wasim**, M. Naseer, S. Khan, and F. Khan, “Method and system for adapting a vision-language machine learning model for image recognition tasks,” U.S. Patent 18541666, Jun. 2025